THE UNIVERSITY OF BRITISH COLUMBIA

CONFINED DIFFUSION OF NANOPARTICLES USING iSCAT MICROSCOPY

by

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Abstract

Particle confinement allows extended measurements of suspended biomolecules and nanoparticles using optical microscopy as it keeps it within the field of view.

However, during confinement, the size-diffusivity relationship changes from that in bulk, where the correction factor is typically assumed without consideration of the occurring measurement conditions such as surface charges and the media.

In this work, we show that by combining interferometric scattering (iSCAT) microscopy with microfluidic nanoparticle confinement we can directly measure the positional distribution and depth-dependent diffusivity. We demonstrate how it depends on particle charge and media properties for both reference beads and lipid nanoparticles. Utilizing range-of-depth motion and in-plane diffusivity as complementary diameter estimates, we showcase how 3D particle tracking under confinement allows for nanoparticle diameter quantification in protein-containing media.

The ability to measure the effect of confined diffusivity improves accuracy in trajectory-based particle sizing in varying media, commonly found in cell biology and nanoparticle sizing.[2].

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Chapter 1

Introduction

1.1 Movivation

- What is the major field i am in? - what are the biological questions - how are they typically slved

- problems with the current solutions

In much of nature, the idea of size scales is widely discussed and investigated. From a basic understanding of physics, one understands that the forces an object feels are directly linked to their mass, or volume and mass density. For example, we can easily understand what will happen to a basketball if we throw it up into the air, as we are sitting on planet earth. However, when we leave the familar size scale, such as having a basketball the size of planet earth, or the size of a gas molecule, our intuition begins to show its cracks.

In size regime of nanoparticles, that is between 20-100nm, many biological particles of utmost importance tend to reside. For instance, in cellular biology, proteins, vesicles, enzymes, and organelles fit between 4 nm to 10 µm. In this size regime, the forces which one feels largely differ from that of say a basketball. Inter-molecular forces between nearby molecules have a strong effect on particle postion, thermal kicks from atomic molecules induce a stochastic motion non-determininstically, and forces of hydrodynamic pressure induce frictions slowing the particle down. Thus, to have any quantitative understnandig we must try to

1.2 Outline for Thesis

- Chapter 1 is an introduction/background - Chpater 2 discusses application of Widefield iSCAT for SPT in the Rayleigh Regime - Conclusion

Chapter 2

SPT in confinement principles

Biology is mostly nanoparticles. Nanoparticles move around stochastically under the rules of brownian motion. To be able to measure the particle's characteristics, one can relate their motion and stochasticity to their properties. This chapter will discuss the phenomena of Brownian motion and how it relates to particle properties, environmental ion concentration, viscosity, hydrodynamic boundaries, and confinement. To directly measure particle position, one can use optical methods, specifically, optical scattering measurements. In this chapter, the theoretical background regarding the particle's stochastic motion is presented, with particular focus on how the processes evolve in confined geometries. There will be a discussion on relating optical signals to their motion, as well as optical methods for particle detection.

Methods that are able to detect single nanoparticles are known as single-molecule measurements. In the field of optical microscopy, single-molecule methods are often referred to as single particle tracking (SPT). This chapter aims to direct the reader through the theoretical context of the work done. First, Brownian motion is discussed, then optical preliminaries for SPT, and lastly, biological motivation is touched upon to continue the introductory words.

2.1 Brownian Motion

The physical phenomenon of diffusion can be rigorously described as the emergent behavior atomic particle undergoing random collisions with members of the surrounding environment. In the microscopic limit, where the radius of a particle is sufficiently small and thermal fluctuations dominate its dynamics, this erratic motion is referred to as Brownian motion [1].

2.1.1 Classical Description of Einstein's Brownian Motion

The classical description of Brownian motion is concerned with how the molecuar-kinetic theory of heat reates with microscopically visible size suspensions in liquid perform movements such

that they can be viewed by a microscope.

2.1.2 Langevin Dynamics

- Brownian motion can alternatively be described using a Langevin Model. - introduce langevin equation - velocity correlation - Arrive at the Fluc-Dissipation - relate it to Einstein's formulation and show that given a stokes drag (sphere)

$$D = \frac{k_b T}{\gamma_0} \tag{2.1}$$

2.1.3 Smoluchowski Equation

- Since Langevin is kind of convient, how do we acctualy link the two methods?

- Fokker plank introdution motivated from the master equation - - ideally summarize van kampen.

- arrive at the general Smoluchowski equation for a random walker - simple plots for descriptive purposes.

2.2 Confined Diffusion Processes

Although diffusion is interesting, there is a wealth of complexity in confinement due to Hydrodynamic effects - quote 111 years here

- outline of this seciton

2.2.1 Hydrodynamic Confinement

- confinemnt from Hydrodynamics - frictional slowing - ...

2.2.2 Geometric Confinement

- i need to check references on if this is relate

2.2.3 Electrostatic Potentials

2.2.4 Example: Polystyrene Bead

2.3 Interferometric Scattering Principles

This section is devoted to the description of physical prinicples associated with light scattering techniques. These techniques are widely used in microscopy, as I have used them in measuring

where my parteicles are. Scattering measurements are of interest as they allow for label-free imaging with a high signal to noise and high temporal resolution.

We begin with the basics of relating light to electromagnetsm, starting with the Maxwell and Helmholtz relations. We continue discussing the scattering of a various size range of spheres, which is the approximation we take in this thesis. We then discuss the various abberations and Considerations before introducting iSCAT microscopy. We tease the experimental apparatus used in tthis work and compare the method to other methods currently used in the field.

2.3.1 Maxwell's Equations and field propagations equations

The photon is the fundamental quanta of the electric field as stated in QFT (cite). It has been shown that because of this, the field equations of light obey the fundamental principle that states

$$I = |\vec{E}(\vec{x}, t)|^2 \tag{2.2}$$

. Thus, it is paramount that we understand how the propagation of the electric fields are described. Luckily for us, this has already been solved in the year 18XX (cite). We present the Maxwell equations in their macroscopic form, which include terms from the material constants.

Definition 1. *Maxwell's Equations in macroscopic form. Let $\vec{x}$ be the position vecotr, t the time scalar. The Maxwell equations are defined as the following set of relations; (CITE ZANGWILL, Scattering small particles, Jackson if needed).*

$$\nabla \cdot \vec{D}(\vec{x}, t) = \rho_f(\vec{x}, t) \tag{2.3}$$

$$\nabla \times \vec{E}(\vec{x}, t) + \frac{\partial \vec{B}(\vec{x}, t)}{\partial t} = 0 \tag{2.4}$$

$$\nabla \cdot \vec{B}(\vec{x}, t) = 0 \tag{2.5}$$

$$\nabla \times \vec{H}(\vec{x}, t) - \frac{\partial \vec{D}(\vec{x}, t)}{\partial t} = \vec{J}_f(\vec{x}, t) \tag{2.6}$$

In addition, for ease of access, the definitions for the Auxillary fields, and scalar potentials are written. We also add the guage invariance;

$$\vec{D}(\vec{x}, t) = \varepsilon_0 \vec{E}(\vec{x}, t) + \vec{P}(\vec{x}, t) \tag{2.7}$$

$$\vec{H}(\vec{x}, t) = \frac{1}{\mu_0} \vec{B}(\vec{x}, t) - \vec{M}(\vec{x}, t) \tag{2.8}$$

To fully describe the material that the field lies in, we also require constitutive relations;

$$\vec{J}_f(\vec{x},t) = \sigma\vec{E}(\vec{x},t) \tag{2.9}$$
$$\vec{B}(\vec{x},t) = \mu\vec{H}(\vec{x},t) \tag{2.10}$$
$$\vec{P}(\vec{x},t) = \varepsilon_0\chi\vec{E}(\vec{x},t) \tag{2.11}$$
$$\tag{2.12}$$

Where the boldface is reserved for tensor quantities.

- List shit that we need to know before calcualting the Scattering from a sphere.

- Helmholtz Equation - Fresnel Equations - Polarization - General Scattering Transfer Matrices

2.3.2 Electromagnetic Scattering of of Homogenous Spheres

To As mentioned, we are tasked with getting the light from the scattered molecules. We approximate them as spheres. Thus, we need to solve the solutions of scattered light from spheres.

We start by using the wave equation for the Electric and Magnetic Fields

$$\nabla^2\vec{E} + k^2\vec{E} = 0, \quad \vec{H} = \frac{1}{i\omega\mu_0}\nabla\times\vec{E} \tag{2.13}$$

The solutions of these quantities will be harmonics of the laplacian in the coordinate basis that we choose. For isotropic scattering, the spherical basis is the one of choice. The solutions to this are known as the Vector Spherical Harmonics. Let ψ be a solution to the scalar wave equation

$$\nabla^2\psi + k^2\psi = 0 \tag{2.14}$$

, then we can form a generator of the general solution by taking the simplest form of solutions and summing all components. Citing Jackson, we find that the solution to this form is given by the seperable form;

$$\begin{aligned}\psi(r,\theta,\phi) &= \Phi(\phi)\Theta(\theta)R(r)\\ \Phi(\phi) &\in \{\cos(m\phi),\sin(m\phi)\}\\ \Theta(\theta) &= P_n^m(\cos\theta)\\ R(r) &\in \{j_n, y_n, j_n + iy_n, j_n - iy_n\}\end{aligned} \tag{2.15}$$

Where j_n, y_n are the spherical Bessel functions, and their complex additions are the hankel functions. Thus, the vector spherical harmonics are best described by seperating these solutions

into even and odd solutions as their prefactor will be cos or sin respectively.

$$\begin{aligned} \vec{M}_{emn} &= \vec{\nabla} \times (\vec{r}\psi_{emn}), \quad \vec{M}_{omn} = \vec{\nabla} \times (\vec{r}\psi_{omn}) \\ \vec{N}_{emn} &= \frac{1}{k}\vec{\nabla} \times \vec{M}_{emn}, \quad \vec{N}_{omn} = \frac{1}{k}\vec{\nabla} \times \vec{M}_{omn} \end{aligned} \tag{2.16}$$

These are found in Eqns 4.17 to 4.20 in Bohren.

The incident light can be assumed to be polarized along the x axis without loss of generality. The general form of these equation will then be the sum of each of the spherical vector harmonics (SVH). subject to the energy conservation boundary condition

$$(\vec{E}_i + \vec{E}_s - \vec{E}_1) \times \hat{r} = (\vec{H}_i + \vec{H}_s - \vec{H}_1) \times \hat{r} = 0 \tag{2.17}$$

Citing Bohren, and stating that using orthongality of the SVHs, we have that the exact solution, which is known as the Mie Theory form of scattering off of a sphere, is given below;

Definition 2. *Mie Theory of scattering off of homogenous spheres*

$$\boxed{\begin{aligned} \vec{E}_i &= E_0 e^{ikr\cos\theta}\hat{x} \\ \vec{E}_s &= \sum_{n=1}^{\infty} E_n(ia_n \vec{N^{(1)}}_{e1n} - b_n \vec{M^{(1)}}_{o1n}) \\ \vec{E}_1 &= \sum_{n=1}^{\infty} E_n(ic_n \vec{M^{(1)}}_{o1n} - d_n \vec{N^{(1)}}_{e1n}) \end{aligned}} \tag{2.18}$$

Since we are only interested in the scattered field, the constants that we care about are written below;

$$\begin{aligned} a_n &= \frac{m\Psi(mx)\Psi'(x) - \Psi(x)\Psi'(mx)}{m\Psi(mx)\xi'(x) - \xi(x)\Psi'(mx)} \\ b_n &= \frac{\Psi(mx)\Psi'(x) - m\Psi(x)\Psi'(mx)}{\Psi(mx)\xi'(x) - m\xi(x)\Psi'(mx)} \end{aligned} \tag{2.19}$$

where Ψ, ξ are the Riccatti-Bessel functions.

In the limit of a small particle from far away, we can mae some cancellations in these constants to simplify the work. Notably, the hankel function approches the limit of a spherical wave,

$$h_n(kr) \xrightarrow{kr \gg 1} (-i)^{n+1}\frac{e^{ikr}}{kr} \tag{2.20}$$

Im this regime, we have that $a_1 \approx \frac{-2i}{3}(ka)^3\frac{m^2-1}{m^2+2}$, and $b_1 \approx 0$. Thus for $n = 1$ we have that

$\pi_1 = 1, \tau_1 = \cos\theta$, thus

$$\begin{aligned} M^{(3)} &\approx (-i)^{n+1}\frac{e^{ikr}}{kr}\left(\cos\phi\pi_n\hat{\theta} - \sin\phi\tau_n\hat{\phi}\right) \\ N^{(3)} &\approx (-i)^{n+1}\frac{e^{ikr}}{kr}\left(\cos\phi\pi_n\hat{\theta} - \sin\phi\tau_n\hat{\phi}\right) \end{aligned} \tag{2.21}$$

can be written in the Rayleigh approximation simply and in the form of the full scattering field;

$$\vec{E}_s = \frac{iE_0e^{ikr}}{kr}\left[\hat{\theta}\cos\phi\sum_{n=1}^{\infty}\frac{2n+1}{n(n+1)}(a_n\tau_n + b_n\pi_n) + \hat{\phi}\sin\phi\sum_{n=1}^{\infty}\frac{2n+1}{n(n+1)}(a_n\pi_n + b_n\tau_n)\right] \tag{2.22}$$

Spoiling the suprise, after plugging in the terms for $n = 1$ we have the following simplification;

$$\begin{aligned} \vec{E}_{s,\perp} &= \vec{E}_{i,\perp}\frac{k^2\alpha e^{ikr}}{r} \\ \vec{E}_{s,\parallel} &= \vec{E}_{i,\parallel}\frac{k^2\alpha e^{ikr}}{r}\cos\theta \end{aligned} \tag{2.23}$$

Then, to find the scattering intensity field, and noting $k = \frac{2\pi}{\lambda/n}$, we have the final scattering field in the Rayleigh regime;

$$\boxed{\vec{E}_s = \vec{E}_0e^{ikr}\frac{\sqrt{8}\alpha\pi^2n_m^2}{\lambda^2r}\sqrt{1+\cos\theta}} \tag{2.24}$$

- Draw some of the iPSFs for Mie theory, from all angles. - manoharan theory and reparam of the main equation

- Set the stage, what do we need to solve, how are we going to do it - Solutions to the wave equation in spherical coordinates - Mie Theory – Examples of Mie spheres – What size range ? - RDG Theory – What size range ?

- Small comparison from far away, medium (mie), and small (RDG), atomic (too small)

2.3.3 Types of Microscopy

Dark field - microscopy where the background is not used widefield - microscopy where the full field of view is used holography - microscopy with a reference field (?)

2.3.4 Scattering Microscopy

As shown, the description of the scattered light from a sphere is full described by Eqn. 2.18, known as the Lorenz-Mie scattering theory (CITE). In the limit of particles on the order of the

size of the wavelength of light, the Rayleigh model is an easier, equivalent method derived from Mie theory.

To collect the light, there have been three main schemes used in literature; forward, side, and backscattering micrpscopy schemes.

As shown in Fig. (reference PSF), the scattering distribution taken at different projections widely varies. In addition, the amount of light that gets collected in each of these regimes also varies.

do i go into the other models?

In the backscattering scheme, the light collected is the sum of the incident light and the scattered light.

$$I(x,y) = |\vec{E}_r + \vec{E}_s|^2 \tag{2.25}$$

given that the incident wave is known as a polarized plane wave, this equation can be rewritten for the scatterd light;

$$I = E_r E_r^* + E_s E_s^* + 2\Re[E_r E_s] \tag{2.26}$$

The incident field will be modulated by a scattering term, which is the Rayleigh scattering amplitude, S, as well as the reference term, R. Thus, the total intensity will be the sum of the scattering amplitude, reference amplitude, and the phase interference between the two fields;

$$I = |E_i|^2 \left[|R|^2 + |S|^2 + 2|R||S|\cos\Delta\varphi\right], \quad \Delta\varphi = \varphi_s\varphi_r \tag{2.27}$$

It should now be noted that the angle of image formation has not been specified. This is of importance as it allows to discuss the various methods of scattering microscopy without loss of generality between methods. Notably, the main difference is the angular content which is obtained from the various scattering models. In the case of spheres, this means using the various angular slices of the resultant scattered field. A full discussion of these comparisons can be found in (CITE).

...

- Olsén's thesis

2.3.5 Interferometric Light Scattering (iSCAT)

The key assumption in iSCAT is the relative magnitude of terms. The reference field is constant. The scattering term $R \approx k^4\alpha^2 = \frac{V^2}{\lambda^4}$. Similarily, the inference term scales as the square root of this. Thus, the scattering term is dominated by the two fields, which we can ignore.

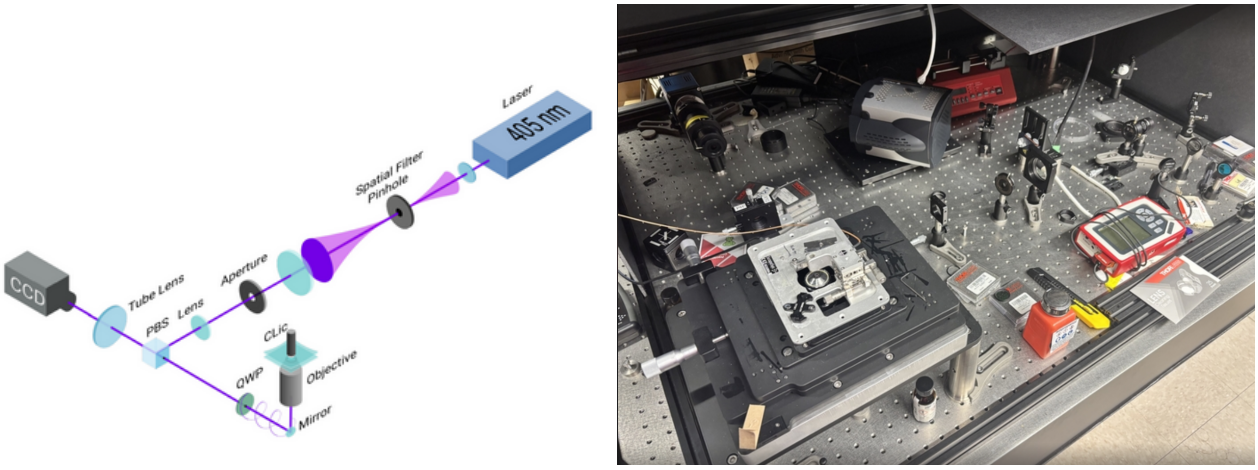


Figure 2.1: **Left**. iSCAT Schematic. **Right**. Image of physical system. A 405 nm laser is sent through a spatial filter, passes through an aperture. The beam is sent through a focusing lens which is reflected off a polarizing beam splitter, polarized through a QWP focused on the back plane of the objective. The scatterd light is then polarized again, passing through the polarizing beam splitter and is focused on the CMOS camera. The final image is collected via the CMOS and saved in binary format.

To accurately measure the positions, we can define the contrast metric, which relates the total field to that of a reference field;

$$\begin{aligned} C &= \frac{I - |E_r|^2}{|E_r|^2} \\ &= \frac{|S|^2 + 2|R||S|\cos\Delta\varphi}{|R|^2} \\ &\xrightarrow{|S|\ll|R|} \frac{2|S|}{|R|}\cos\Delta\varphi \end{aligned} \tag{2.28}$$

The phase of the interference consists of a sum of terms consisting of phases accumulated through the traversal of the scattered field. The total phase is dominated by the Gouy phase, which consists of phase accumulated when light is passed through a converging and diverging field, the defocus phase, which can be parameterized to a relative phase shift, as well as the phase accumulated phase difference, which is typical of a Michaelson-Interferomer type system.

Experimentally, this system is described in the set of schematics in Fig. 2.1.

2.3.6 Resolution limits of optical microscopy

The last thing to touch upon is understanding how the image that is being enlarged arrives to the camera, which consists of discussion of optical aberrations.

As mentioned in the formulation for the electric field scatter we have that any optical field consists of a set of plane waves that construct and deconstruct to form a field. That is, the

electric field lies in the basis of a fourier series of terms of complex exponentials. This fact allows one to describe the propagation of a wave in throughout z.

$$\vec{E}(x,y,z) = \iint_{\Omega} \vec{E}_0(k_x,k_y) e^{-i\left(xk_x+yk_y+\sqrt{k^2-k_x^2-k_y^2}\right)} dk_x dk_y, \quad \Omega = \{x \in \mathbb{R} : x^2 \leq k^2\} \tag{2.29}$$

We have that the energy of the wave is fixed by the trigonometric relation of the transverse wave vectors. Notably, this is fixed by the relation $|k| = \frac{2\pi}{\lambda}$.

The propagated wave passes through an objective which can be approximated by a lens. Using the ABCD matrix formulation, we can find that to have a clear image, the imaging condition must be satisfied, that is

$$\frac{1}{z_1} + \frac{1}{z_2} = \frac{1}{f} \tag{2.30}$$

where z_1, z_2 are the positions of the observer and the object, and f is the focal distance of the lens. Thus, the magnification of the lens is defined as $M = z_2/z_1$. However, to understand the aberrations it is nessecary to have a field formulation of this problem. The simplest way to do this is described using linear optics theory.

Briefly, we define a pupil function $P(x,y)$ which is unitary if $r \leq R$, where R is the radius of lens. The point spread function of the object is then defined as a delta function through a two lens system. This is given by the fourier transform of the point source masked by the pupil function;

$$H(x,y) = \iint_{\Omega} P(x',y') e^{\frac{ik}{2}\left(\frac{1}{z_1}-\frac{1}{f_1}-\frac{1}{f_2}+\frac{1}{z_2}\right)\sqrt{x'^2}+y'^2} e^{-\frac{ik}{z_2}(xx'+yy')} dx'dy' \tag{2.31}$$

Expanding this, one can use a zero-order Hankel transform to integrate over the radius and find that the point-spread function, H, is defined in terms of the NA and the magnification;

$$H(r,z) = \int_0^1 e^{-\frac{ik}{2}\frac{\mathrm{NA}^2}{n}\rho^2 z} J_0\left(\frac{k\mathrm{NA}}{M}\rho r\right)\rho d\rho \tag{2.32}$$

where ρ is the rescaled radial vector $\rho = r'/R$, $\mathrm{NA} = nR/f_1$, and $M = nf_2/f_1$.

In perfect focus, we have that $z = 0$, for $z \neq 0$ we have a defocusing aberration which is defined as the secondary exponential. For a point source, which is diffraction limited, we find that the following is the limit of the optical microscope;

$$H(r,0) = \left|\frac{J_1(k\mathrm{NA}r)}{k\mathrm{NA}r}\right|^2 \tag{2.33}$$

We can solve that the minimum distance between the peak of the PSF and the minumum is solved by $k\mathrm{NA}r = x_0$ where x_0 is the first root of $J_1(x)$. This is classical known as the minimum transverse imaging criterion;

$$r_{\min} = 0.61\frac{\lambda}{\mathrm{NA}} \tag{2.34}$$

Similar results are found for the minimum axial imaging criterion by setting $r = 0$ and integrating.

$$z_{\min} = \frac{2\lambda n}{\mathrm{NA}^2} \tag{2.35}$$

At this point, we invite the fact that not all the light that scatters passes through teh lens. Specifically, the range of k_x and k_y are fundamentally limited by the wavelength. The resolution limit is λ/NA. In the case of scattering microscopy, this is not a large problem as the scatteirng pattern is much larger, hwoever, in the limit of two particle near each other, the spatial reoslution limit must be enforced.

discuss how we go from reality (focus plane) to image (image plane).

- sphereical abberations - guoy phase ...

- final kind of check

theortical 100 nm particle -¿ scattering field -¿ abberations

2.4 In-Vivo Considerations

- Now that we more or less know how BM works, why is it important in cells?

2.4.1 why does this matter for biological applications

2.4.2 Examples of Brownian Motion in Biology

2.5 Brownian motion as a method for measurements of size in biology

2.5.1 survey of methods using in this field

- Optical Methods - Bulk: DLS, - Single Molecule: SPT, NTA, FCS,

2.5.2 how confinemen effects these measurements

- Literature on why these methods are lacking

2.5.3 Probing of surface interactions with these methods

Chapter 3

Widefield backscattering iscat for SPT in 3d confinement tracking and analysis

3.1 Questions

3.2 Rayleigh scattering of nanoparticles

3.2.1 Parameterization used in this thesis

3.3 Methods

3.3.1 Optical System

3.3.2 Convex-lens induced confinement

3.4 Results i guess

3.5 Discussion of Results

Chapter 4

Conclusion

4.1 Conclusion

4.2 Future outlook

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